Task 3: Adversity Prediction Model

Model Choice and Rationale:

I have trained one gradient-boosted-tree classifier, specifically the model HistGradientBoostingClassifier from scikit-learn per horizon $\tau \in \{5,...,30\}$, each predicting P(trade adverse at τ) where the label is 1[side·(Mt-TP)<0]. (here 1 is the indicator function). Boosted trees are chosen because the adversity signal is weak, non-linear and interaction-heavy (client × volume × volatility). ed trees are chosen because the adversity signal is weak, non-linear and interaction-heavy (client × volume × volatility); trees capture this without manual feature crossing, not much scaling, and their probability outputs can be used directly into the Task-4 threshold rule. Data is split 60/20/20 by date (chronological) to prevent look-ahead leakage; rolling features are shifted so each trade sees only strictly prior information.

Feature selection and justification:

Sixteen strictly-causal features span three groups.

- Order characteristics: client one-hots (toxicity differs sharply by client, Task 1), side, volume and log-volume (large prints are more informed), the order-book spread and relative spread.
- Execution geometry: TP - M0 and the signed offset side · (TP-M0), capturing where the print landed inside the book (the single most informative input)
- Market State: realized volatility (std of the last 20 mid-returns) and short-term momentum (sum of the last 5 returns), plus time-of-day fraction. These let the model condition on the regime. We deliberately excluded future mids (the target) and any non-causal aggregate.

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Task 3 - Feature vector ordering (index : name)
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 0 : client_A
 1 : client_B
 2 : client_C
 3 : client_D
 4 : client_E
 5 : client_F
 6 : side
 7 : volume
 8 : log_volume
 9 : spread
10 : rel_spread
11 : tp_minus_mid
12 : signed_offset
13 : tod_frac
14 : realized_vol
15 : momentum
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Performance (metrics averaged over the six horizons) is shown below. Test accuracy 0.558 and log-loss 0.684 beat the 0.5 / 0.693 random baselines, confirming a genuine edge (even though the accuracy is not that high). Short-horizon mid prediction is near the efficiency limit, so the value is in ranking trades by toxicity rather than perfect classification which is why we do not actually worry about the modest accuracy scores. Train and test metrics are close, indicating little overfitting (l2 regularisation, min 200 samples/leaf). Recall is deliberately low at the 0.5 cut because most flow is benign; Task 4 re-optimises the operating threshold for PnL rather than accuracy.

split	accuracy	precision	recall	log_loss
train	0.567143	0.570177	0.243819	0.679881
validation	0.555884	0.528737	0.233074	0.685028
test	0.558488	0.519436	0.233271	0.684289